

CREATIVE INSTITUTE DATA SCIENCE

Data Cleaning

Impute Missing Value using scikit learn

- Numerical Variable
- Categorical variable

Different Strategy for different variable

- Simpleimputer
- Pipeline
- ColumnTransformer

Now import library & load dataset

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.impute import SimpleImputer
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
```

now loading the dataset & check the shape of dataset

```
train=pd.read_csv('train.csv')
test=pd.read_csv('test.csv')
print('train shape',train.shape)
print('test shape',test.shape)
```

now convert data into x train & ytrain & x-test data

```
X_train=train.drop(columns='SalePrice',axis=1)
y_train=train['SalePrice']
X_test=test.copy()
print('train shape',X_train.shape)
print('test shape',y_train.shape)
print('X_test Shape',X_test.shape)
```

now find the total missing value present in dataset

```
null_var=X_train.isnull().sum()
null_var
```

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now find the numerical missing column

find the only numerical value

```
num_col=X_train.select_dtypes(include=['int64','float64']).columns
num_col
```

now find missing value columns name in numerical variable

```
miss_num_col=[i for i in num_col if null_var[i]>0]
miss_num_col
```

there is 3 columns in missing value in numerical variable

now find categorical variable who have missing value present

```
cat_col=X_train.select_dtypes(include="O").columns
cat_col
```

now find any missing value in this columns

```
miss_cat_col=[i for i in cat_col if null_var[i]>0]
miss_cat_col
```

now we will use different different method to fill value

if we want to fill some value using mean in some columns, some value using mode, median in some columns & some value fill using constant value .. use different method to fill missing value

now create list of where put missing value

```
num_col_mean=['LotFrontage']
num_col_median=['MasVnrArea', 'GarageYrBlt']
cat_col_mode=['Alley', 'MasVnrType', 'BsmtQual', 'BsmtCond', 'BsmtExposure', 'BsmtFinType1',
              'BsmtFinType2', 'Electrical', 'FireplaceQu']

cat_col_missing=['GarageType', 'GarageFinish', 'GarageQual', 'GarageCond', 'PoolQC',
                 'Fence', 'MiscFeature']
```

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for use this list for different technique create pipeline

```
num_col_mean_imputer=Pipeline(steps=[("imputer",SimpleImputer(strategy="mean"))])
num_col_median_imputer=Pipeline(steps=[("imputer",SimpleImputer(strategy="median"))])
cat_col_mode_imputer=Pipeline(steps=[("imputer",SimpleImputer(strategy="most_frequent"))])
cat_col_missing_imputer=Pipeline(steps=[("imputer",SimpleImputer(strategy="constant",fill_value="missing"))])
```

we have created pipeline using different strategy

now apply this strategy on different variable use columntransformer

```
preprocessor=ColumnTransformer(transformers=[('mean_imputer',num_col_mean_imputer,num_col_mean),
                                             ('median_imputer',num_col_median_imputer,num_col_median),
                                             ('mode_imputer',cat_col_mode_imputer,cat_col_mode),
                                             ('missing_imputer',cat_col_missing_imputer,cat_col_missing)])
```

Now we have created transformer for different strategy on different value

Now we apply this strategy on dataset

```
preprocessor.fit(X_train)
```

if you want full information use

```
preprocessor.transform
```

if you want check which value has given fit()

```
preprocessor.named_transformers_["mean_imputer"].named_steps['imputer'].statistics_
```

now you can check. Which value we imputed.

```
preprocessor.named_transformers_["mean_imputer"].named_steps['imputer'].statistics_
```

you can apply for different method. For imputer the value

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now we have all the value and apply or transform all the value in dataset

```
X_train_clean=preprocessor.transform(X_train)
X_train_clean
```

Now we have 2dArray value

Now we check which columns has missing value to fill

```
preprocessor.transformers_
```

use this we seen there is some columns which is drop by default. Whose doesn't have missing values on original dataset

now we need to create dataframe

keep in mind that we use ditto 2 ditto sequence which was use in transformer

```
X_train_clean_miss=pd.DataFrame(X_train_clean,
                                columns=num_col_mean+num_col_median+cat_col_mode+cat_col_missing)
X_train_clean_miss.head()
```

Now we have cleaned dataset..

If you want to find there is any missing value

```
X_train_clean_miss.isnull().sum().sum()
```

There isnt any missing value in any columns